Muhammad Usman Aziz

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PROFESSIONAL SUMMARY

Results-oriented AI and Full-Stack Software Engineer with 1+ years of enterprise experience developing scalable ERPNext solutions, full-stack web applications, and AI-driven workflows. Proven background in designing secure document verification systems, automated ETL migration pipelines, and modern web architectures (React, TanStack Start, Node.js). Backed by a strong Computer Science degree from FAST-NUCES and 7+ years of technical mentorship experience.

EDUCATION

Bachelor of Computer Science, FAST-NUCES, CFD Campus — Sep 2021 - June 2025

SKILLS

Programming Languages	Python, TypeScript, JavaScript, C++, SQL, R, HTML5/CSS3
AI & Machine Learning	Generative AI, LLMs, RAG, Fine-Tuning (LoRA/PEFT), NLP, Deep Learning, PyTorch, TensorFlow, OpenCV, Keras, Transformers, Pandas, NumPy, Scikit-learn
Full-Stack & Web	React.js, TanStack Start, Node.js, Express.js, Frappe / ERPNext, Tailwind CSS, REST APIs, Webhooks, Dexie (IndexedDB), Cloudflare Workers
Mobile Development	React Native, Capacitor, Android Studio, Expo Go
Databases & ERP	MariaDB, MySQL, MongoDB, ERPNext, Google Sheets API, QuickBooks Integration
Developer Tools	Git, GitHub, VS Code, Cursor, Postman, Vite, Jupyter Notebook, Colab, PyCharm

EXPERIENCE

Software Engineer — Oct 2025 - Present
Agrovisions — *Faisalabad, Punjab, Pakistan*

- Developed and customized enterprise ERPNext modules, automating sales and invoicing workflows that reduced manual processing time by 40% and minimized reconciliation errors.
- Engineered tamper-proof cryptographic document verification with dynamic QR codes, securing 500+ commercial invoices and certificates against falsification.
- Spearheaded automated ETL pipelines to extract, sanitize, and migrate 5+ years of legacy QuickBooks Desktop accounting records across 20+ transaction types into ERPNext with 100% data fidelity.
- Built and maintained full-stack web applications and REST API microservices handling live bidirectional inventory synchronization and automated webhook dispatching.

Computer Science & STEM Instructor — Aug 2019 - Present
The Century Academy of Science Commerce and Arts — *Faisalabad, Punjab, Pakistan*

- Instructed and mentored 150+ high school and intermediate students across 7+ years in Computer Science, Mathematics, and Physics, achieving a 95%+ board exam pass rate.
- Coached students in fundamental data structures, algorithmic thinking, and Python programming, with multiple mentees securing admissions into top-tier university CS programs.

Software & AI Internships — July 2024 - Sep 2024
CodeAlpha • Oasis Infobyte • GrowIntern — *Remote*

- **CodeAlpha (AI Intern):** Developed an AI-powered autonomous delivery robot system in Python with dynamic obstacle pathfinding.
- **Oasis Infobyte (Data Analyst Intern):** Conducted exploratory data analysis (EDA) and K-means customer segmentation on retail sales datasets.

- **GrowIntern (ML Intern):** Applied machine learning algorithms to real-world datasets with hands-on model training and evaluation.

PROJECTS

M Medium Technical Blogs

Deaf-ED (Final Year Project) - (MERN, OpenCV, and LSTM)

- Designed an AI-powered assistive mobile app using PSL animations and computer vision (LSTM/OpenCV), achieving 90%+ real-time sign recognition accuracy for inclusive early childhood education.

Stockly - (TanStack Start, React, TypeScript, Tailwind CSS, Dexie, Capacitor)

- Developed an offline-first retail inventory and profit manager with IndexedDB persistence, sub-second query latency, and automated OAuth sync to Google Sheets for multi-device management.

QuickBooks Desktop Data Extractor - (Python, QBXML SDK, VBScript COM, Web UI)

- Engineered a high-throughput SDK extraction engine using QBXML and COM Bridge, exporting 20+ transaction types and master records into structured JSON with real-time Web UI configuration.

QuickBooks to ERPNext Migration Engine - (Frappe, ERPNext, Python, Pandas)

- Built an automated ETL data pipeline and custom Frappe application to transform, validate, and load 5+ years of QuickBooks accounting, payroll, and stock entries into ERPNext DocTypes.

ERPNext Document Verification Portal - (ERPNext, Frappe, Python, JavaScript, QR Code)

- Built a cryptographic QR verification portal for ERPNext, enabling instant single-scan validation and eliminating invoice tampering risks across hundreds of commercial records.

Gemini Web 2 API - (TypeScript, Node.js, REST API)

- Engineered an API gateway exposing web-based Gemini AI capabilities as OpenAI-compatible REST endpoints with session management and multi-turn conversational support.

Medical AI Assistant - (RAG, LLMs, Vector Retrieval)

- Engineered an intelligent clinical RAG assistant integrating vector search and LLMs to cross-reference diagnostic medical literature and synthesize structured clinical summaries.

Oxigen E-Commerce Platform - (React, Node.js, Express, ERPNext REST API, Tailwind CSS)

- Built a high-performance retail e-commerce platform and admin portal integrated with ERPNext, supporting real-time multi-warehouse stock deduction and automated sales invoicing.

Shopify AI Assistant - (Node.js, Express, LLM API, Shopify Storefront API)

- Built an intelligent AI chatbot for Shopify storefronts handling automated 24/7 customer support, semantic product search, and live order status inquiries.

Shayar AI - (LSTM, GRU, RNN, NLP)

- Trained deep recurrent neural networks (LSTM/GRU) on extensive classical poetic corpora to generate metrically structured Urdu couplets with interactive lexical prompt steering.